SIMATS ENGINEERING
ASSESSMENT TOOL 1

NAME: Siva Dakshitha K
REG.NO: 192425395
COURSE NAME: CSA1603
COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding ***(2.5)***	Constraint Analysis ***(2.5)***	Solution Development ***(2.5)***	Application of Concepts ***(1.5)***	Justification ***(1)***	***Total(10)***
Weightage	*25%*	25%	25%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I __Siva Dakshitha K __ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.
Siva Dakshitha K
Signature of the student:

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

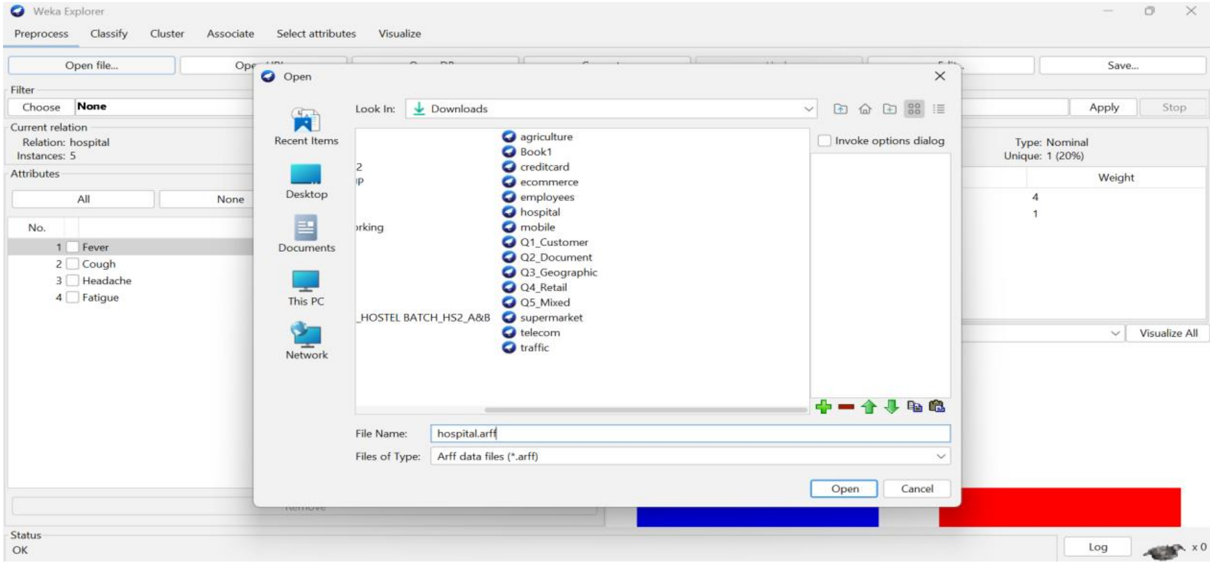
Constraint-Based Problem Solving (6 Marks) (BL4–BL5) – Reframed with New Scenarios

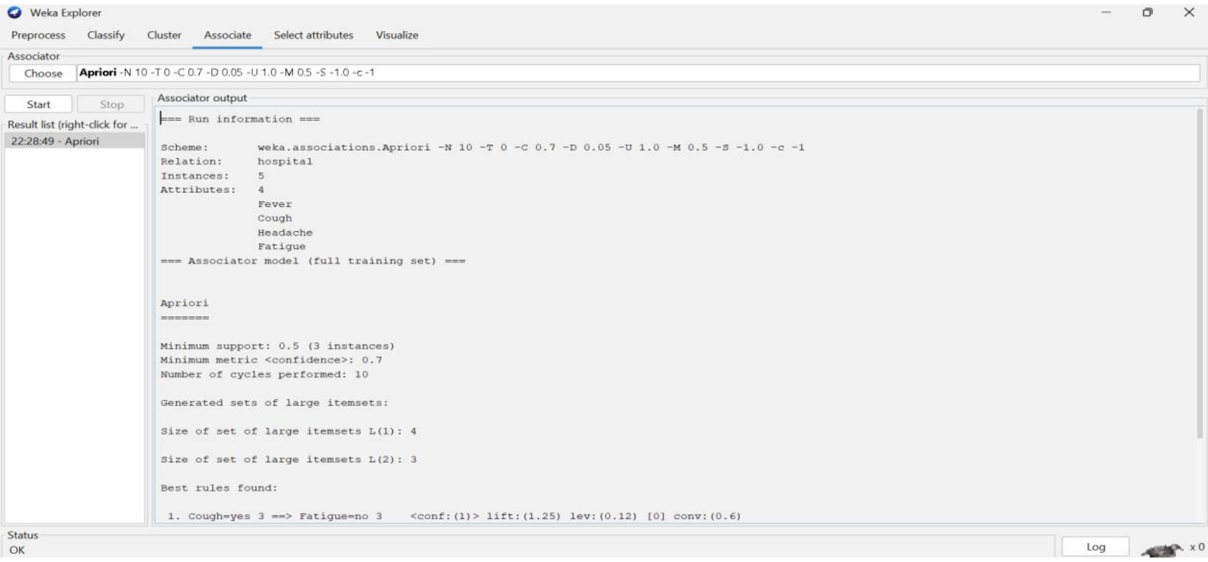
Q1. Dataset: Hospital Patient Records

PID	**Symptoms Observed**
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:
Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints. *(BL4 – 1.2 Marks)*

ANSWER:


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Save...
Open
Look In:
Downloads
Recent Items
Desktop
Documents
This PC
Network
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
Q2_Document
Q3_Geographic
Q4_Retail
Q5_Mixed
supermarket
telecom
traffic
Invoke options dialog
File Name:
hospital.arff
Files of Type:
Arff data files (*.arff)
Open
Cancel
Current relation
Relation: hospital
Instances: 5
Attributes
No.
1 Fever
2 Cough
3 Headache
4 Fatigue
Type: Nominal
Unique: 1 (20%)
Weight
Visualize All
Status
OK
Log


Weka Explorer
Associator
Choose
Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
22:28:49 - Apriori
Associator output
=== Run information ===
Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1
Relation: hospital
Instances: 5
Attributes: 4
Fever
Cough
Headache
Fatigue
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.5 (3 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 10
Generated sets of large itemsets:
Size of set of large itemsets L(1): 4
Size of set of large itemsets L(2): 3
Best rules found:
1. Cough=yes 3 ==> Fatigue=no 3 <conf:(1)> lift:(1.25) lev:(0.12) [0] conv:(0.6)
Status
OK
Log


Weka Explorer
Associator
Choose
Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
22:28:49 - Apriori
Associator output
Cough
Headache
Fatigue
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.5 (3 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 10
Generated sets of large itemsets:
Size of set of large itemsets L(1): 4
Size of set of large itemsets L(2): 3
Best rules found:
1. Cough=yes 3 ==> Fatigue=no 3 <conf:(1)> lift:(1.25) lev:(0.12) [0] conv:(0.6)
2. Headache=yes 3 ==> Fatigue=no 3 <conf:(1)> lift:(1.25) lev:(0.12) [0] conv:(0.6)
3. Fatigue=no 4 ==> Fever=yes 3 <conf:(0.75)> lift:(0.94) lev:(-0.04) [0] conv:(0.4)
4. Fever=yes 4 ==> Fatigue=no 3 <conf:(0.75)> lift:(0.94) lev:(-0.04) [0] conv:(0.4)
5. Fatigue=no 4 ==> Cough=yes 3 <conf:(0.75)> lift:(1.25) lev:(0.12) [0] conv:(0.8)
6. Fatigue=no 4 ==> Headache=yes 3 <conf:(0.75)> lift:(1.25) lev:(0.12) [0] conv:(0.8)
Status
OK
Log

Conclusion

The Fever-only itemset is frequent, but there is no meaningful Fever-containing association rule satisfying both support ≥ 50% and confidence ≥ 70% from this dataset.

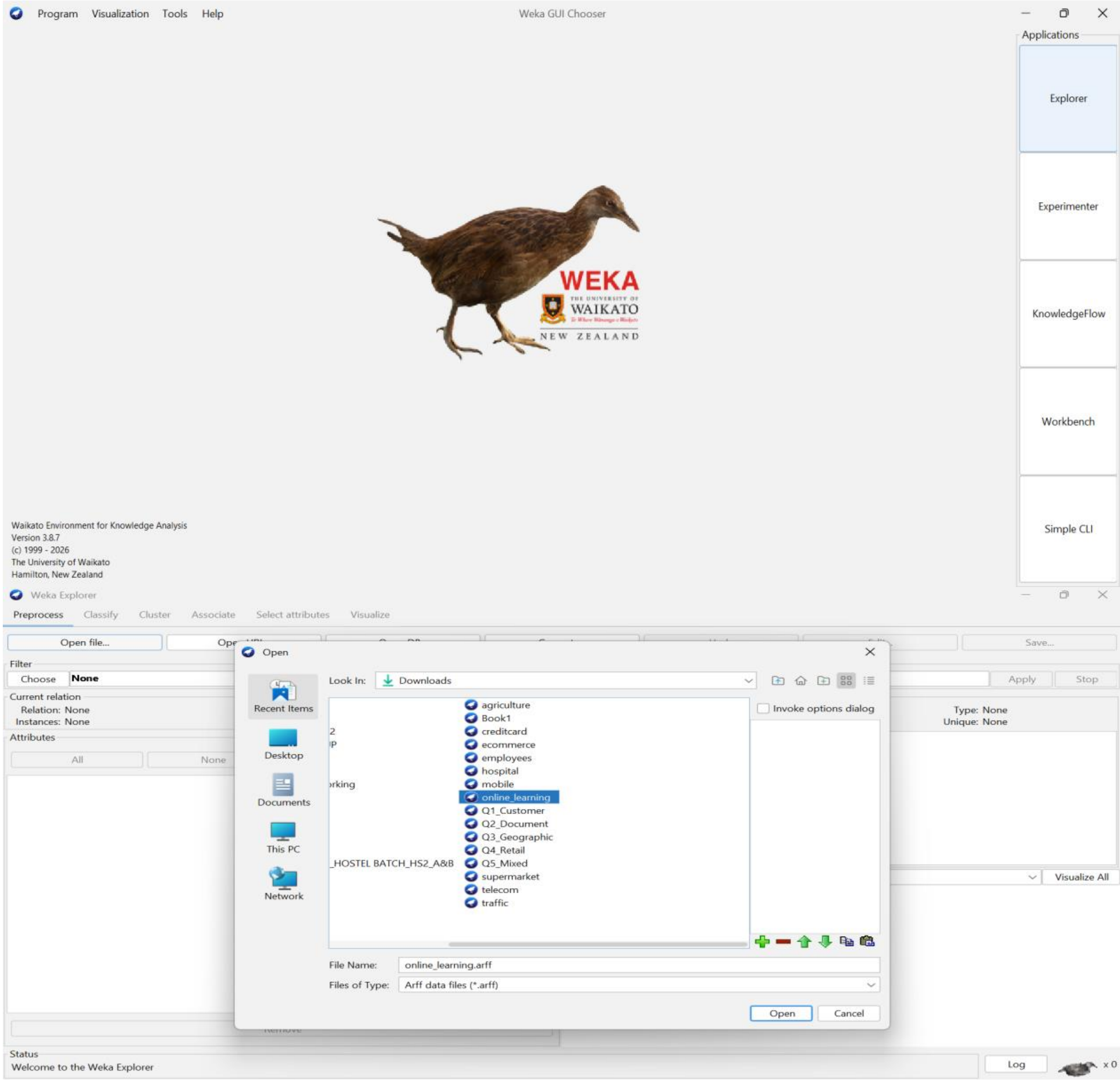
This is an important observation for your answer: WEKA may produce no valid Fever-based rule under these exact constraints.

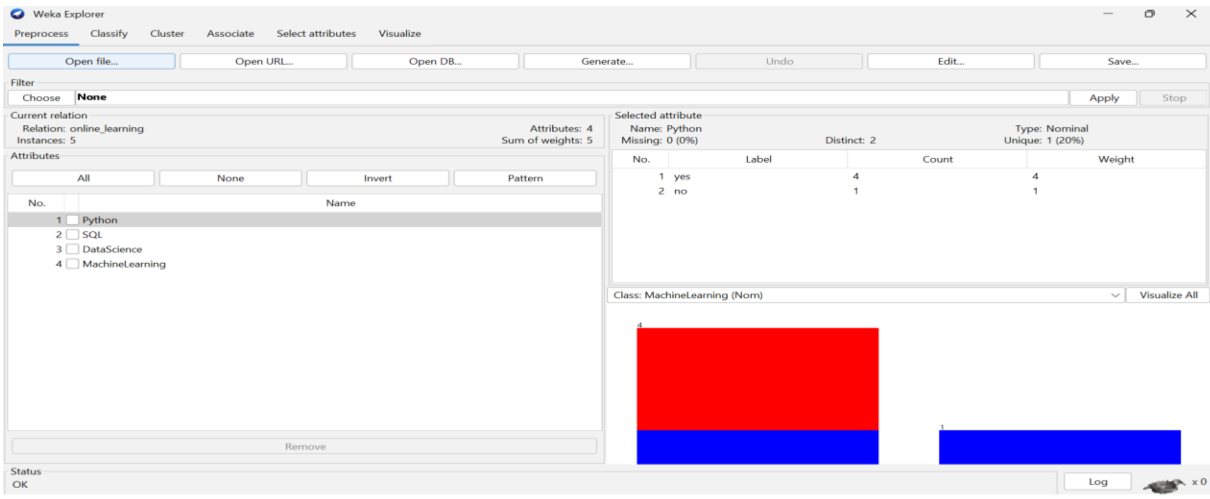
Q2. Dataset: Online Learning Platform

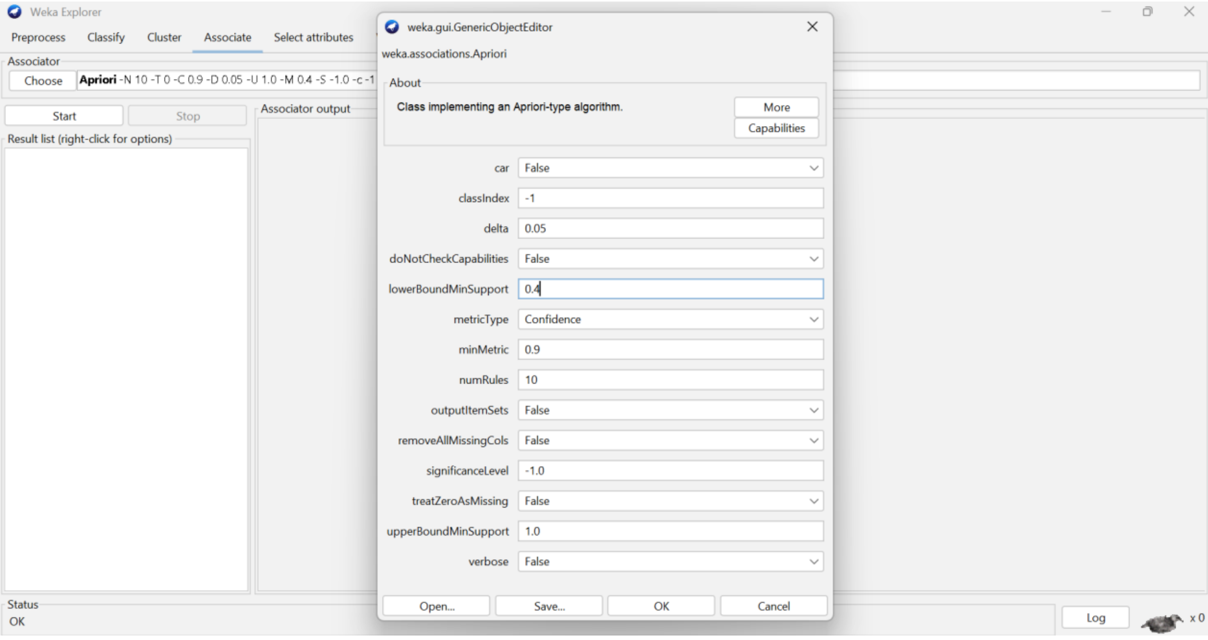
Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:
Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining. *(BL4 – 1.2 Marks)*

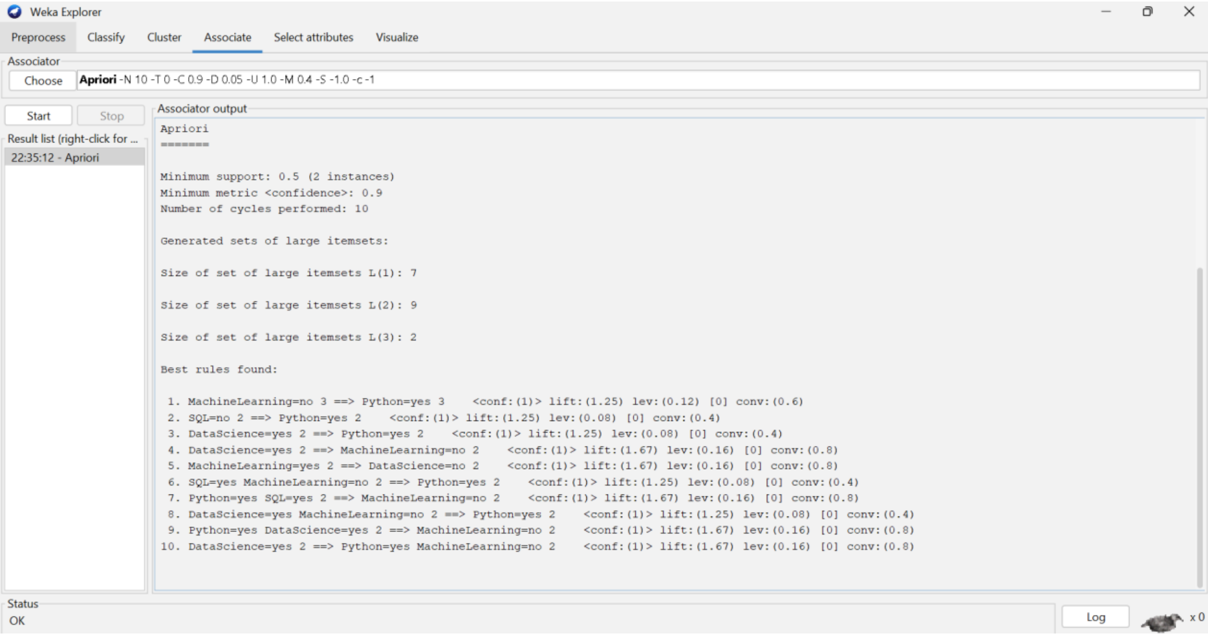
ANSWER:


Program Visualization Tools Help
Weka GUI Chooser
Applications
Explorer
Experimenter
KnowledgeFlow
Workbench
Simple CLI
WEKA
THE UNIVERSITY OF WAIKATO
NEW ZEALAND
Waikato Environment for Knowledge Analysis
Version 3.8.7
(c) 1999 - 2026
The University of Waikato
Hamilton, New Zealand
Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Open file...
Save...
Filter
Choose None
Apply Stop
Current relation
Relation: None
Instances: None
Attributes
All None
Type: None
Unique: None
Visualize All
Open
Look In: Downloads
Recent Items
Desktop
Documents
This PC
Network
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
online_learning
Q1_Customer
Q2_Document
Q3_Geographic
Q4_Retail
Q5_Mixed
supermarket
telecom
traffic
_HOSTEL BATCH_HS2_A&B
Invoke options dialog
File Name: online_learning.arff
Files of Type: Arff data files (*.arff)
Open Cancel
Status
Welcome to the Weka Explorer
Log
x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Open file... Open URL... Open DB... Generate... Undo Edit... Save...
Filter
Choose None
Apply Stop
Current relation
Relation: online_learning
Instances: 5
Attributes: 4
Sum of weights: 5
Attributes
All None Invert Pattern
No. Name
1 Python
2 SQL
3 DataScience
4 MachineLearning
Remove
Selected attribute
Name: Python
Missing: 0 (0%)
Distinct: 2
Type: Nominal
Unique: 1 (20%)
No. Label Count Weight
1 yes 4 4
2 no 1 1
Class: MachineLearning (Nom)
Visualize All
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Associator
Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Start
Stop
Associator output
Result list (right-click for options)
weka.gui.GenericObjectEditor
weka.associations.Apriori
About
Class implementing an Apriori-type algorithm.
More
Capabilities
car False
classIndex -1
delta 0.05
doNotCheckCapabilities False
lowerBoundMinSupport 0.4
metricType Confidence
minMetric 0.9
numRules 10
outputItemSets False
removeAllMissingCols False
significanceLevel -1.0
treatZeroAsMissing False
upperBoundMinSupport 1.0
verbose False
Open...
Save...
OK
Cancel
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
22:35:12 - Apriori
Associator output
=== Run information ===
Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Relation: online_learning
Instances: 5
Attributes: 4
Python
SQL
DataScience
MachineLearning
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.5 (2 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 10
Generated sets of large itemsets:
Size of set of large itemsets L(1): 7
Size of set of large itemsets L(2): 9
Size of set of large itemsets L(3): 2
Best rules found:
Status
OK
Log
x 0

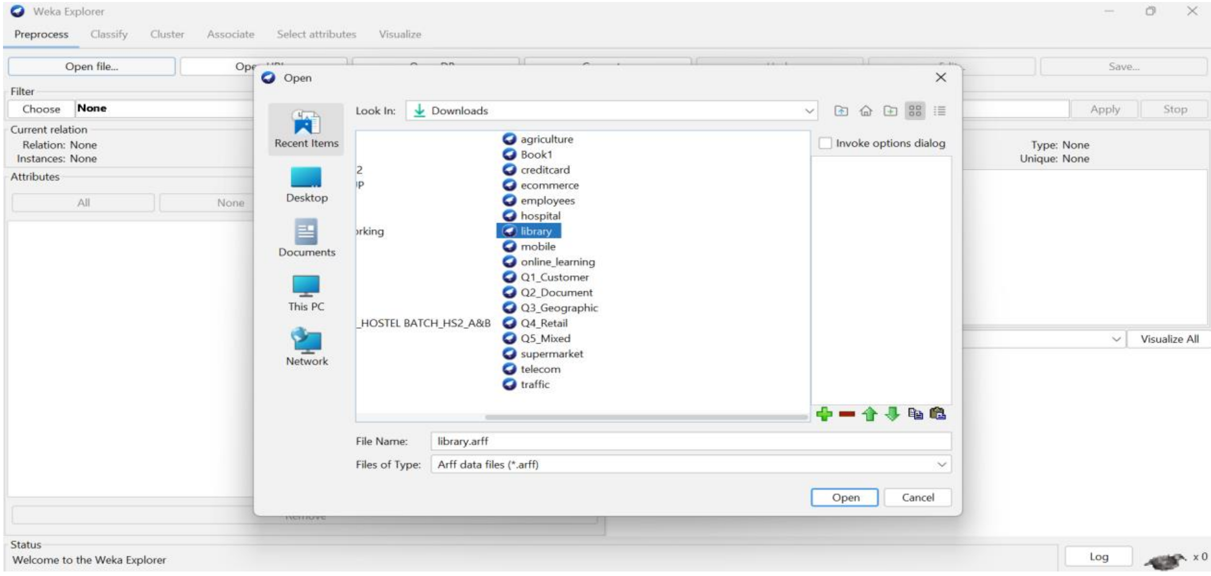


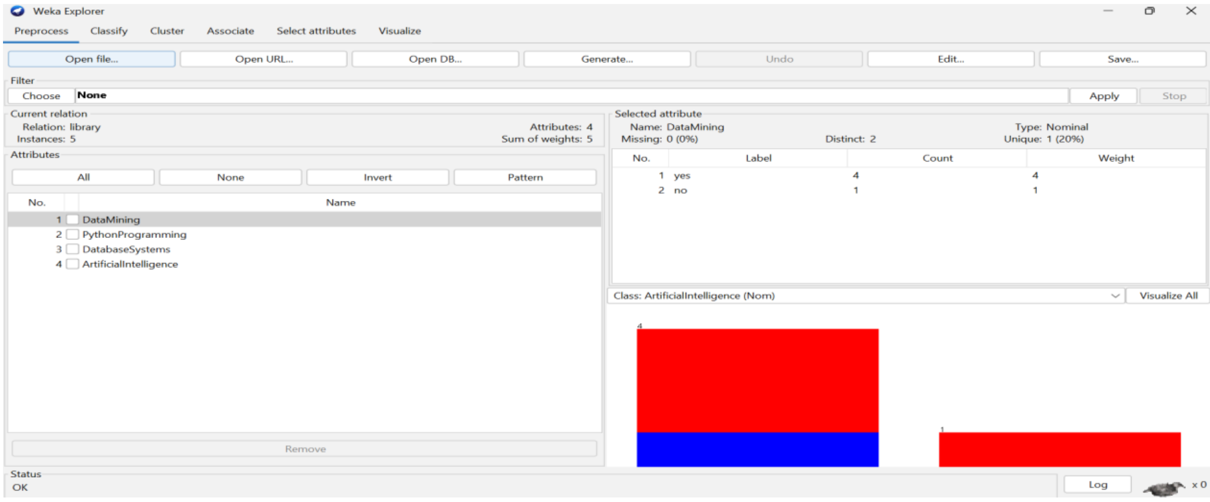
Q3. Dataset: Library Borrowing Records

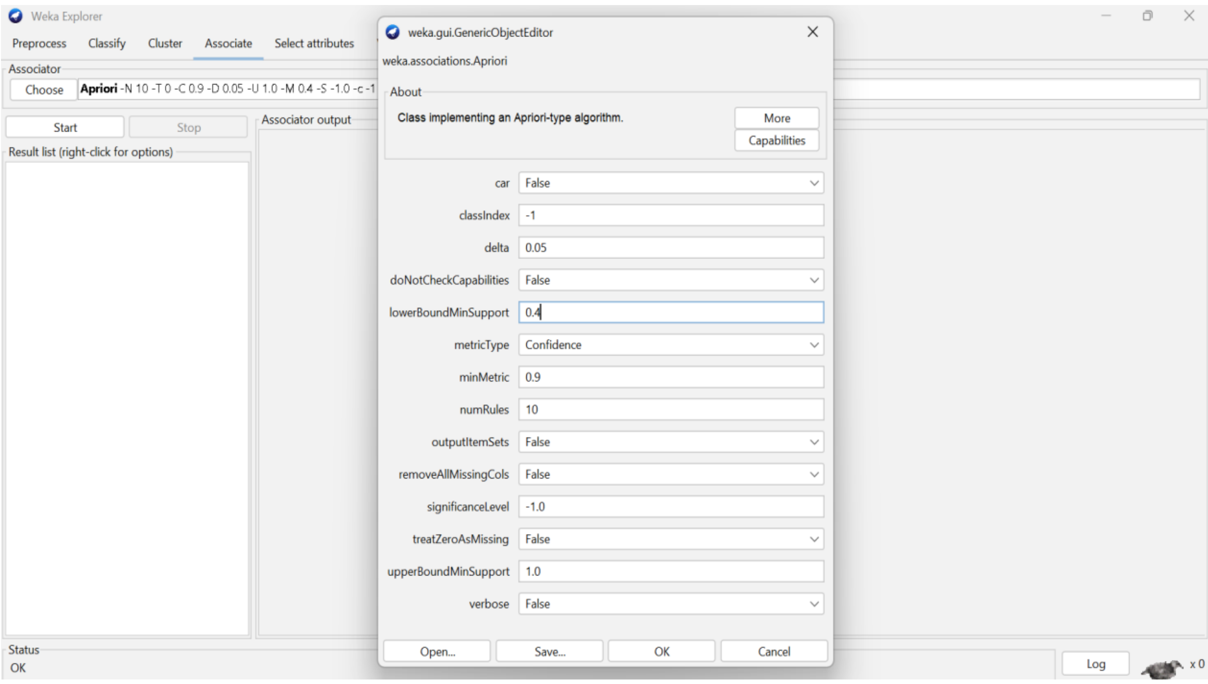
Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

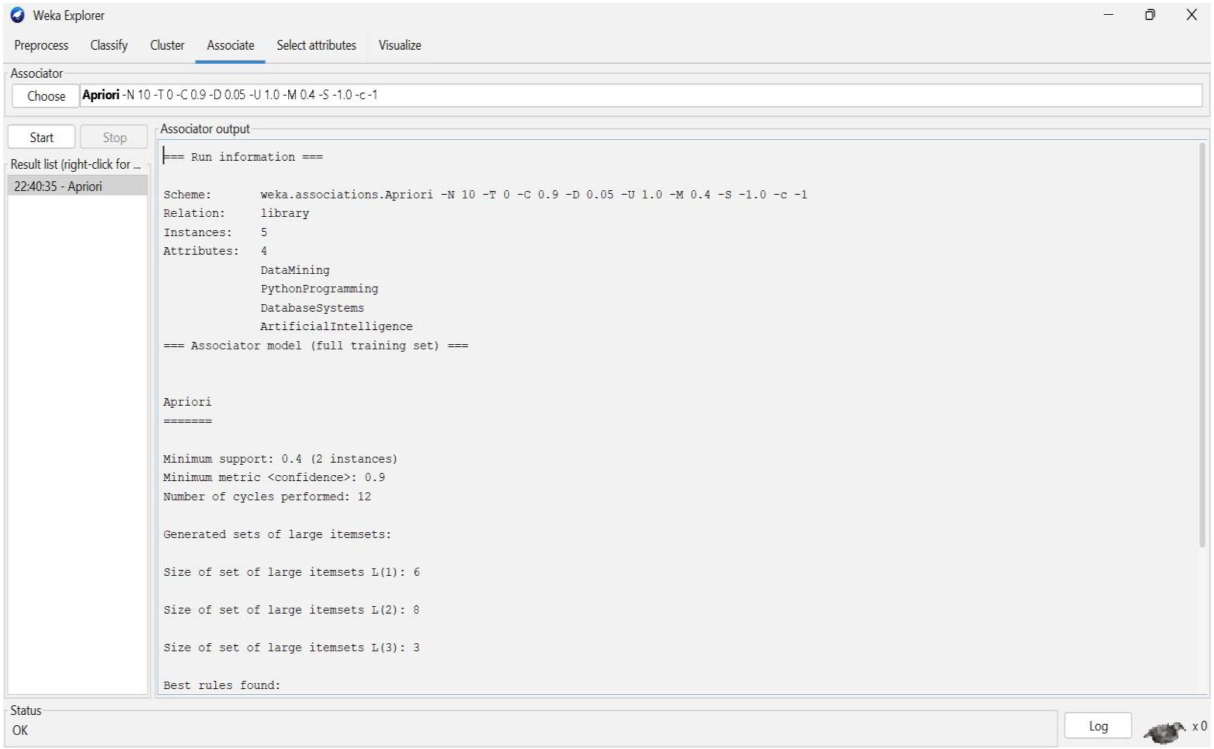
Question:
Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation. *(BL5 – 1.2 Marks)*

ANSWER:


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open
Look In: Downloads
Recent Items
Desktop
Documents
This PC
Network
agriculture
Book1
creditcard
ecommerce
employees
hospital
library
mobile
online_learning
Q1_Customer
Q2_Document
Q3_Geographic
Q4_Retail
Q5_Mixed
supermarket
telecom
traffic
Invoke options dialog
File Name: library.arff
Files of Type: Arff data files (*.arff)
Open
Cancel
Status
Welcome to the Weka Explorer
Log


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: library
Instances: 5
Attributes: 4
Sum of weights: 5
Selected attribute
Name: DataMining
Missing: 0 (0%)
Distinct: 2
Type: Nominal
Unique: 1 (20%)
No. Label Count Weight
1 yes 4 4
2 no 1 1
Attributes
All
None
Invert
Pattern
No. Name
1 DataMining
2 PythonProgramming
3 DatabaseSystems
4 ArtificialIntelligence
Class: ArtificialIntelligence (Nom)
Visualize All
Remove
Status
OK
Log


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Associator
Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Start
Stop
Associator output
Result list (right-click for options)
weka.gui.GenericObjectEditor
weka.associations.Apriori
About
Class implementing an Apriori-type algorithm.
More
Capabilities
car False
classIndex -1
delta 0.05
doNotCheckCapabilities False
lowerBoundMinSupport 0.4
metricType Confidence
minMetric 0.9
numRules 10
outputItemSets False
removeAllMissingCols False
significanceLevel -1.0
treatZeroAsMissing False
upperBoundMinSupport 1.0
verbose False
Open...
Save...
OK
Cancel
Status
OK
Log



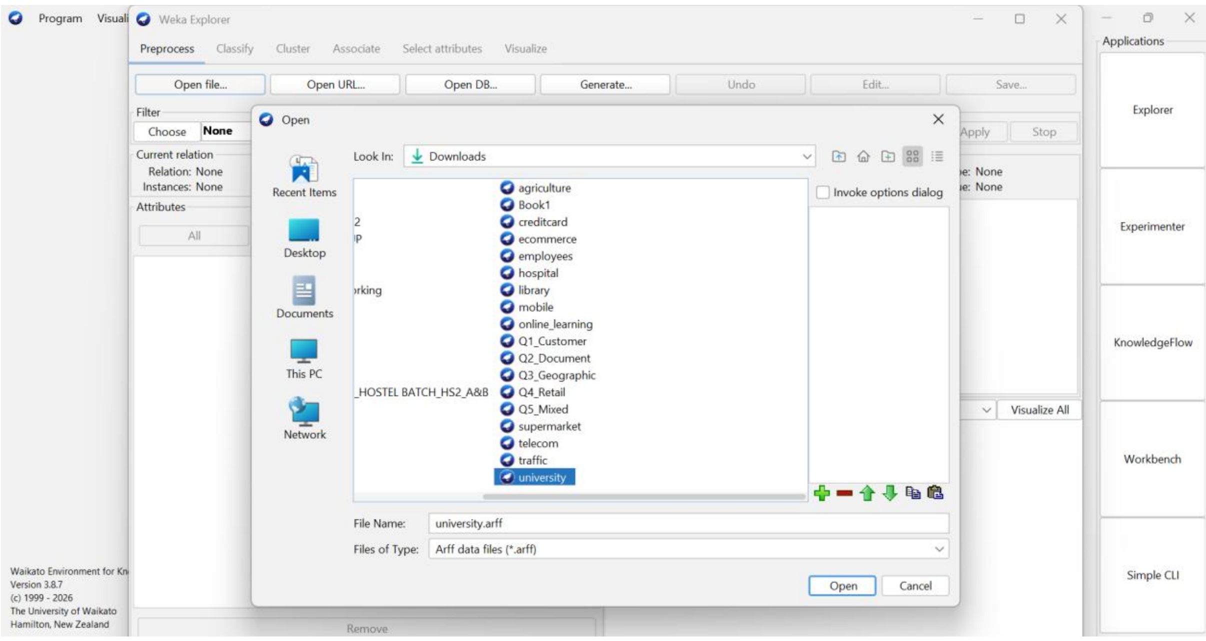
Q4. Dataset: University Course Hierarchy

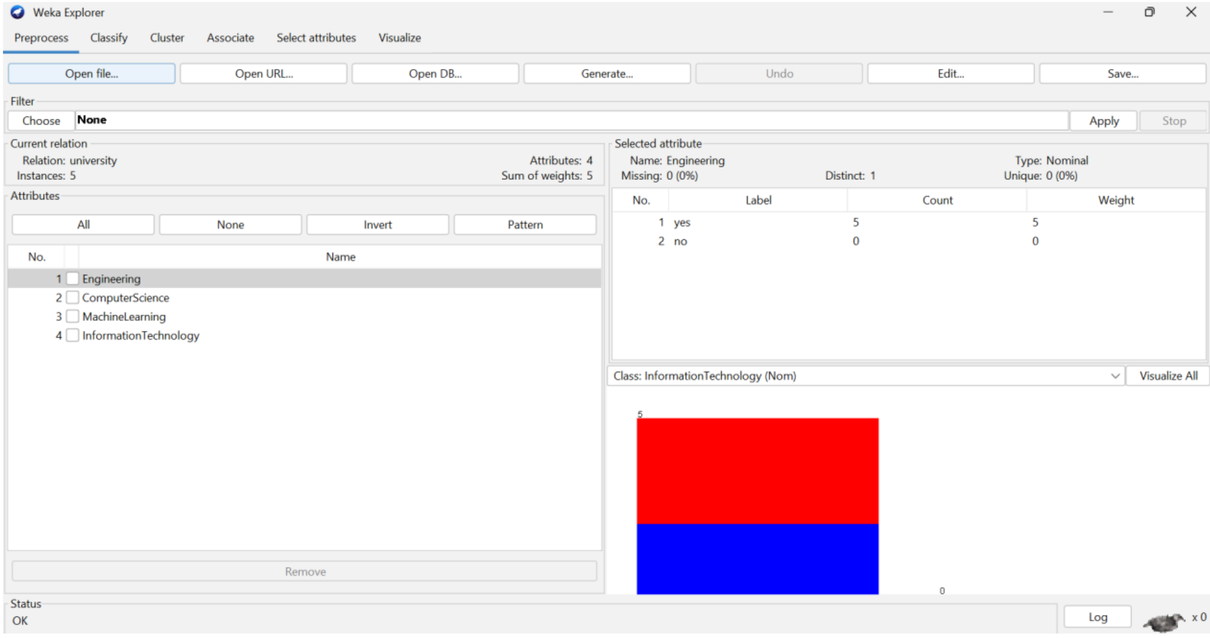
Student ID	**Courses Taken**
S1	Engineering, Computer Science, Machine Learning

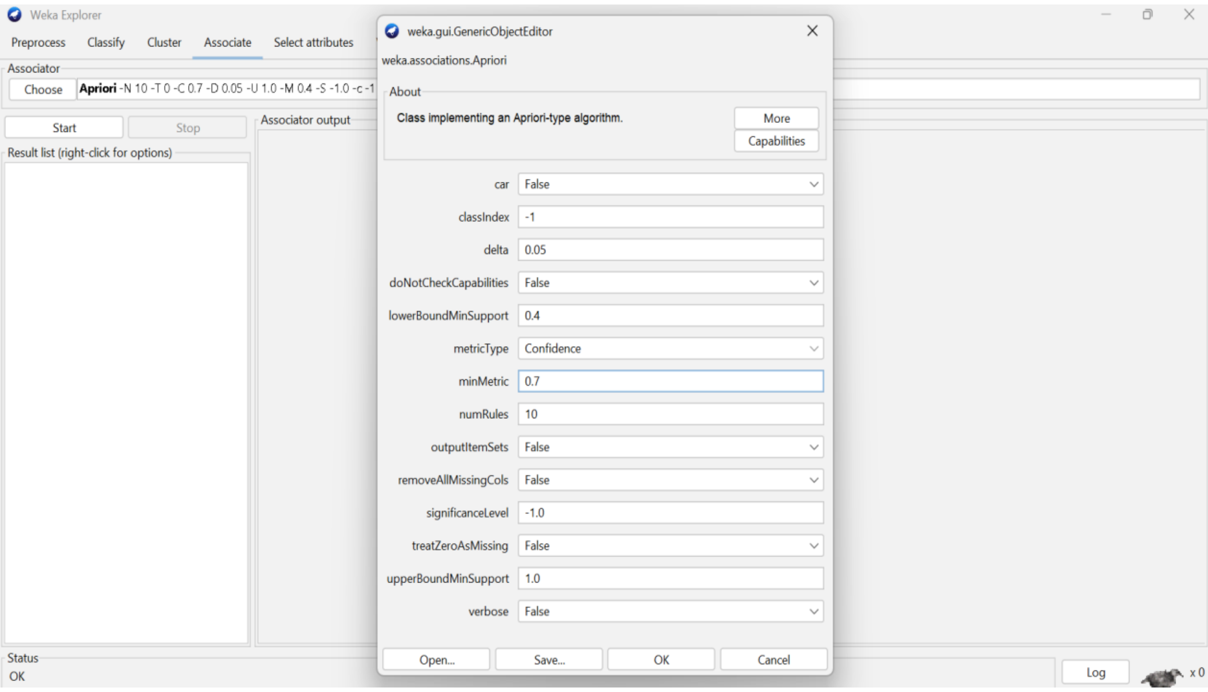
Student ID	**Courses Taken**
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

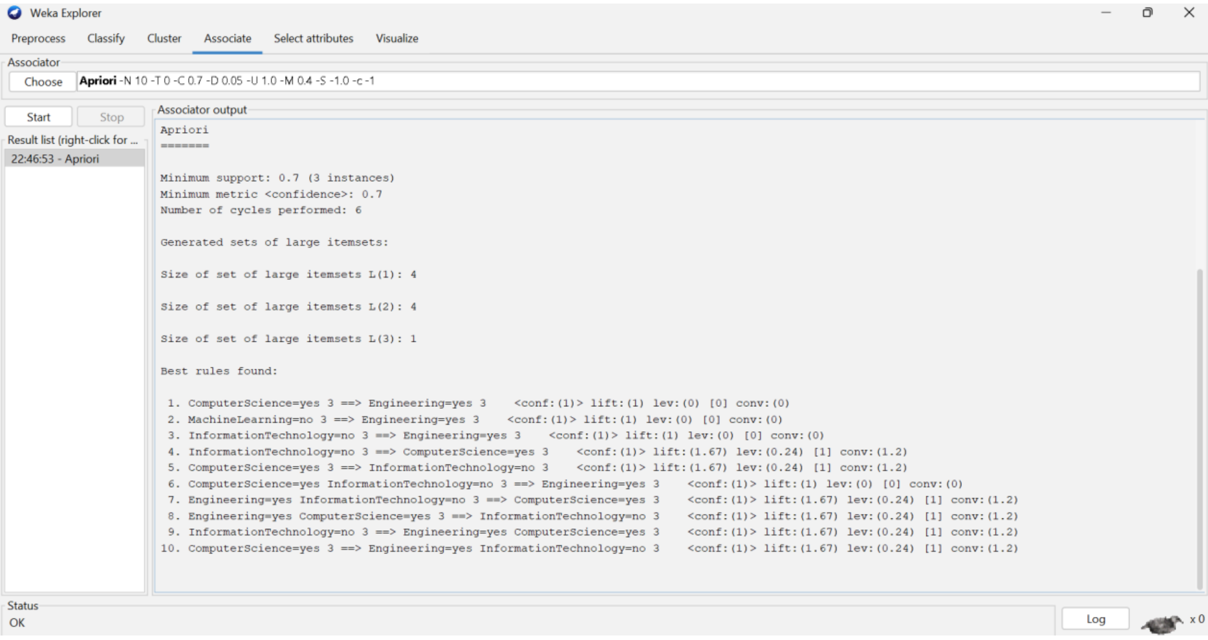
Question:
Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules. *(BL5 – 1.2 Marks)*

ANSWER:




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: university
Instances: 5
Attributes: 4
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No.
Name
1 Engineering
2 ComputerScience
3 MachineLearning
4 InformationTechnology
Remove
Selected attribute
Name: Engineering
Missing: 0 (0%)
Distinct: 1
Type: Nominal
Unique: 0 (0%)
No.
Label
Count
Weight
1 yes 5 5
2 no 0 0
Class: InformationTechnology (Nom)
Visualize All
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Associator
Choose
Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Start
Stop
Associator output
Result list (right-click for options)
weka.gui.GenericObjectEditor
weka.associations.Apriori
About
Class implementing an Apriori-type algorithm.
More
Capabilities
car False
classIndex -1
delta 0.05
doNotCheckCapabilities False
lowerBoundMinSupport 0.4
metricType Confidence
minMetric 0.7
numRules 10
outputItemSets False
removeAllMissingCols False
significanceLevel -1.0
treatZeroAsMissing False
upperBoundMinSupport 1.0
verbose False
Open...
Save...
OK
Cancel
Status
OK
Log
x 0



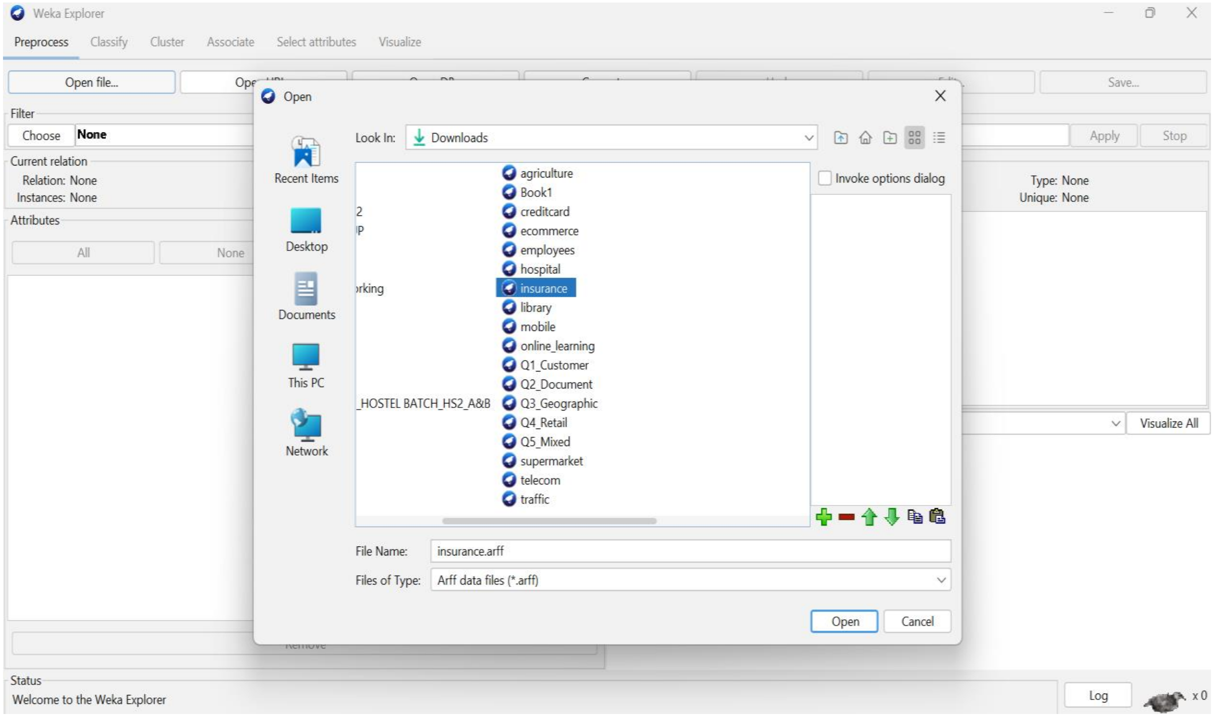
Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

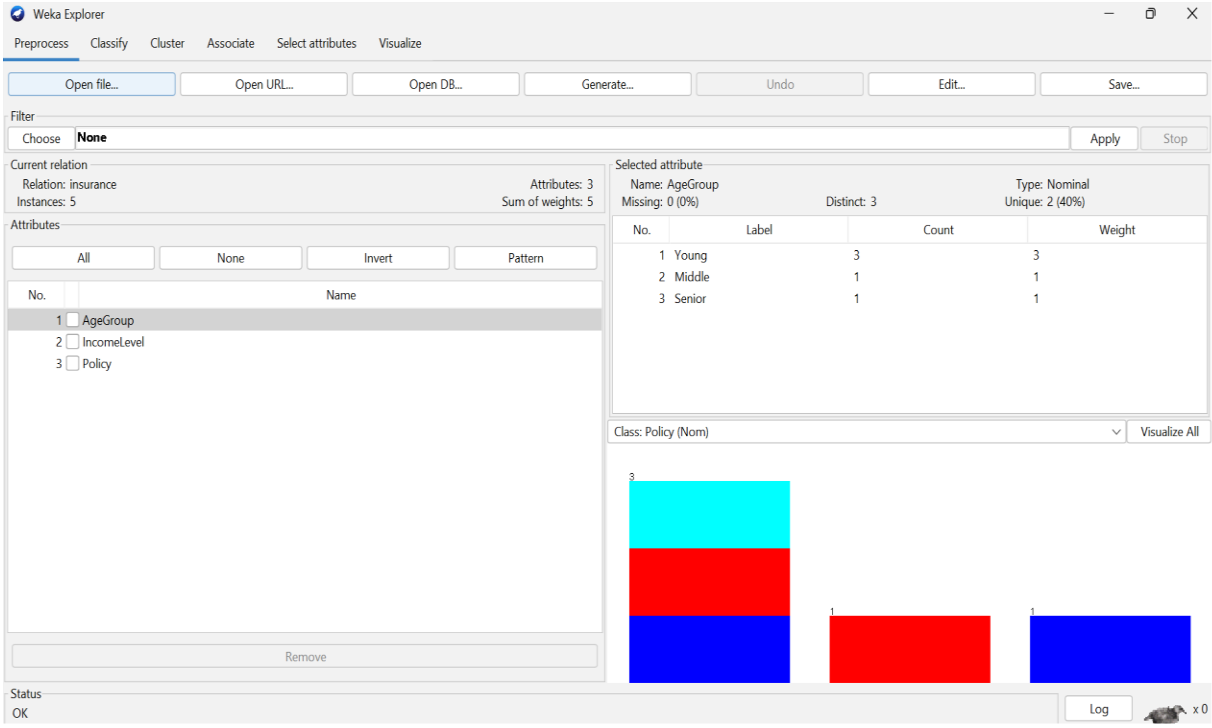
Question:
Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies. *(BL4 – 1.2 Marks)*

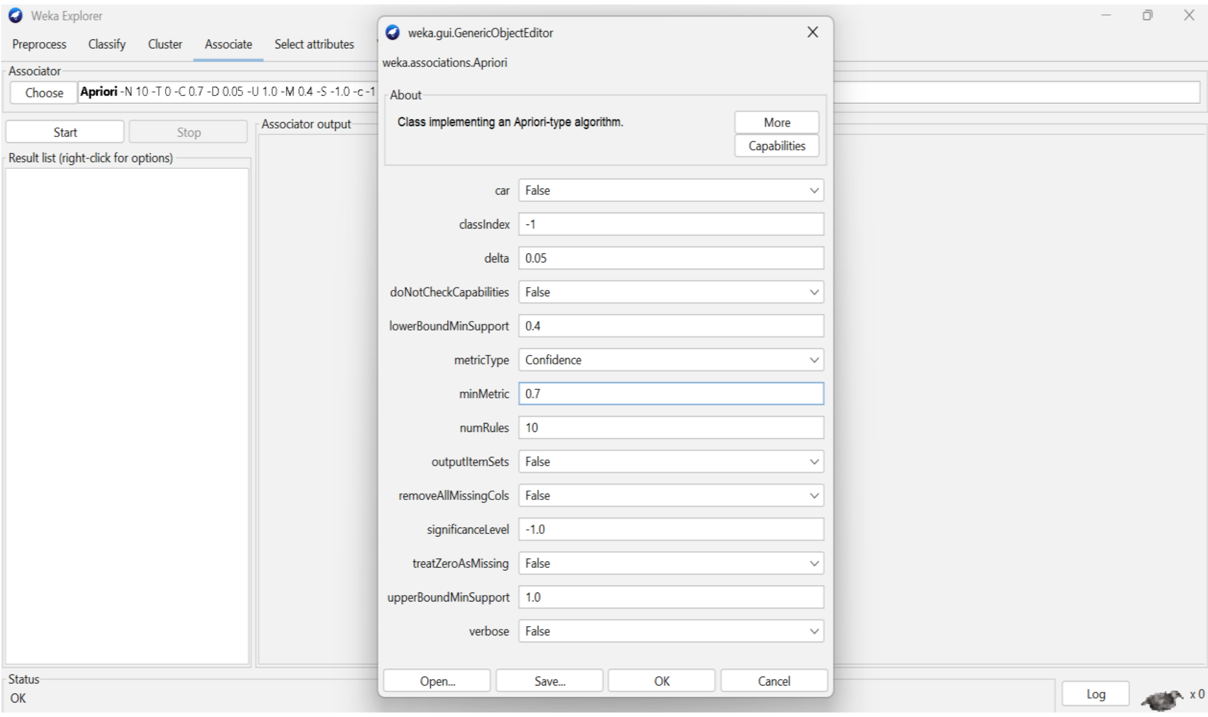
ANSWER:

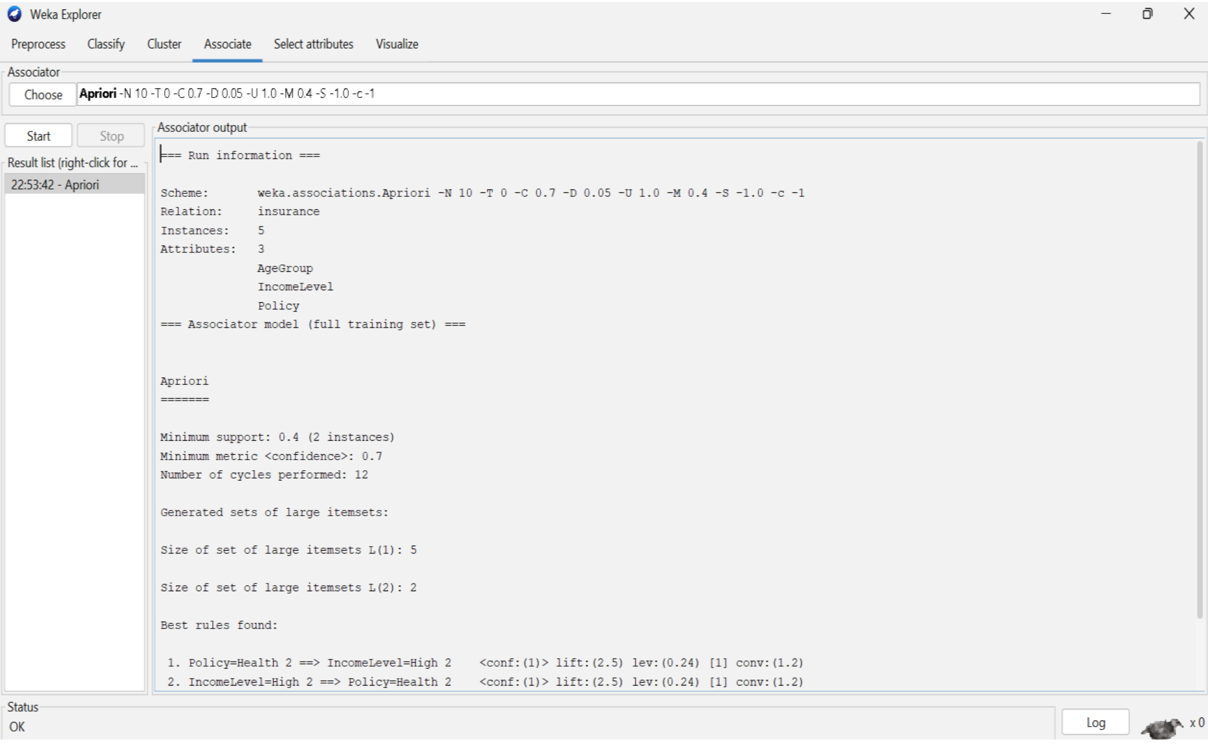
Program
Visualization
Tools
Help
Weka GUI Chooser
Applications
Explorer
Experimenter
KnowledgeFlow
Workbench
Simple CLI
WEKA
Waikato Environment for Knowledge Analysis
Version 3.8.7
(c) 1999 - 2026
The University of Waikato
Hamilton, New Zealand

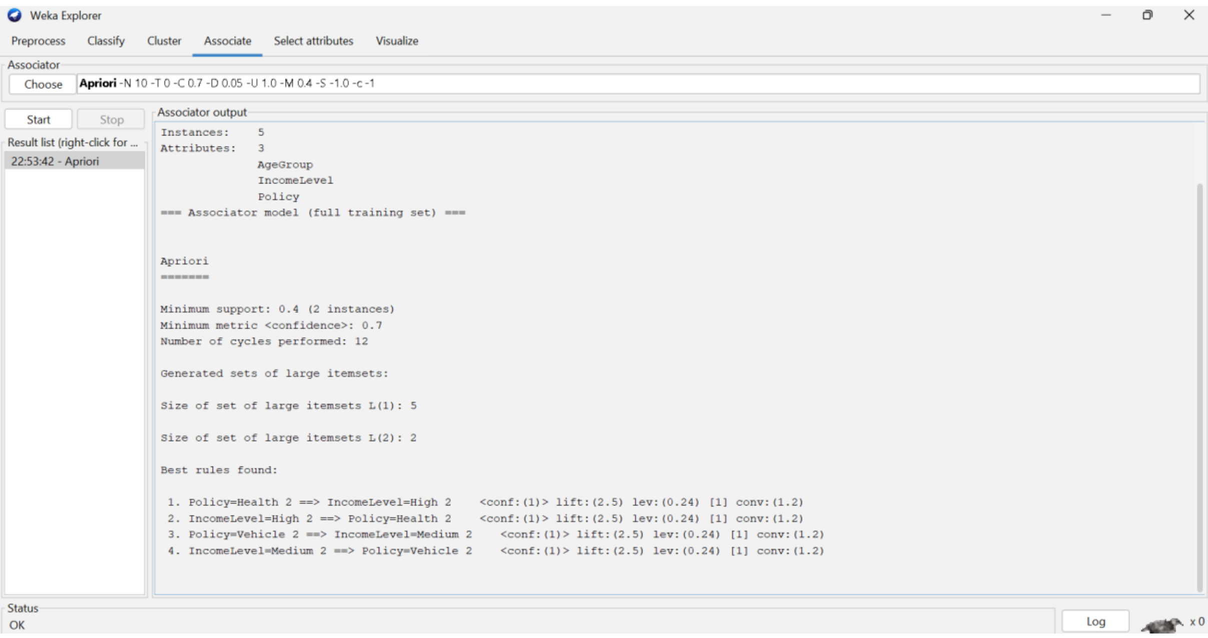


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: None
Instances: None
Attributes
All
None
Type: None
Unique: None
Visualize All
Open
Look In:
Downloads
Recent Items
Desktop
Documents
This PC
Network
Invoke options dialog
agriculture
Book1
creditcard
ecommerce
employees
hospital
insurance
library
mobile
online_learning
Q1_Customer
Q2_Document
Q3_Geographic
Q4_Retail
Q5_Mixed
supermarket
telecom
traffic
File Name:
insurance.arff
Files of Type:
Arff data files (*.arff)
Open
Cancel
Status
Welcome to the Weka Explorer
Log
x 0






Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
22:53:42 - Apriori
Associator output
=== Run information ===
Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Relation: insurance
Instances: 5
Attributes: 3
AgeGroup
IncomeLevel
Policy
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.4 (2 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 12
Generated sets of large itemsets:
Size of set of large itemsets L(1): 5
Size of set of large itemsets L(2): 2
Best rules found:
1. Policy=Health 2 ==> IncomeLevel=High 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
2. IncomeLevel=High 2 ==> Policy=Health 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
22:53:42 - Apriori
Associator output
Instances: 5
Attributes: 3
AgeGroup
IncomeLevel
Policy
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.4 (2 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 12
Generated sets of large itemsets:
Size of set of large itemsets L(1): 5
Size of set of large itemsets L(2): 2
Best rules found:
1. Policy=Health 2 ==> IncomeLevel=High 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
2. IncomeLevel=High 2 ==> Policy=Health 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
3. Policy=Vehicle 2 ==> IncomeLevel=Medium 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
4. IncomeLevel=Medium 2 ==> Policy=Vehicle 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
Status
OK
Log
x 0